

Администрирование локальных сетей

Лабораторная работа №4

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Цель

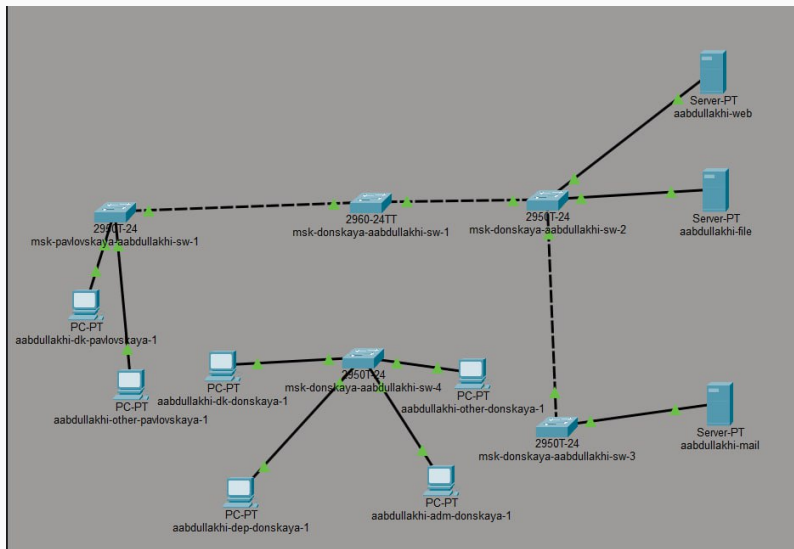
Выполнить базовую настройку сетевого оборудования: присвоение имён, настройка IP-адресов на VLAN, организация защищённого удалённого доступа по протоколу SSH и подготовка коммутаторов к работе в корпоративной сети.

Выполнение лабораторной работы

В среде Cisco Packet Tracer была смоделирована сеть в соответствии с топологией L1. В состав схемы вошли следующие коммутаторы:

- `msk-pavlovskaya-aabdullakhi-sw-1`
- `msk-donskaya-aabdullakhi-sw-1`
- `msk-donskaya-aabdullakhi-sw-2`
- `msk-donskaya-aabdullakhi-sw-3`
- `msk-donskaya-aabdullakhi-sw-4`

Кроме того, в сети присутствуют оконечные устройства: рабочие станции (dk, dep, adm, other), а также серверы web, file и mail. Соединения выполнены через интерфейсы FastEthernet в соответствии с заданием.



На каждом устройстве были выполнены следующие действия: - создан и активирован интерфейс VLAN 2; - назначен IP-адрес из диапазона управления; - прописан шлюз по умолчанию; - настроены пароли на console и VTY; - установлен пароль enable secret; - включено шифрование всех паролей (service password-encryption); - создан локальный пользователь **admin**; - задано доменное имя **donskaya.rudn.edu**; - сгенерированы RSA-ключи; - разрешён доступ только по SSH на виртуальных терминалах; - выполнено сохранение конфигурации.

Настройка msk-donskaya-aabdullakhi-sw-1

```
msk-donskaya-aabdullakhi-sw-1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Switch(config)#hostname msk-donskaya-aabdullakhi-sw-1
msk-donskaya-aabdullakhi-sw-1(config)#interface vlan2
msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

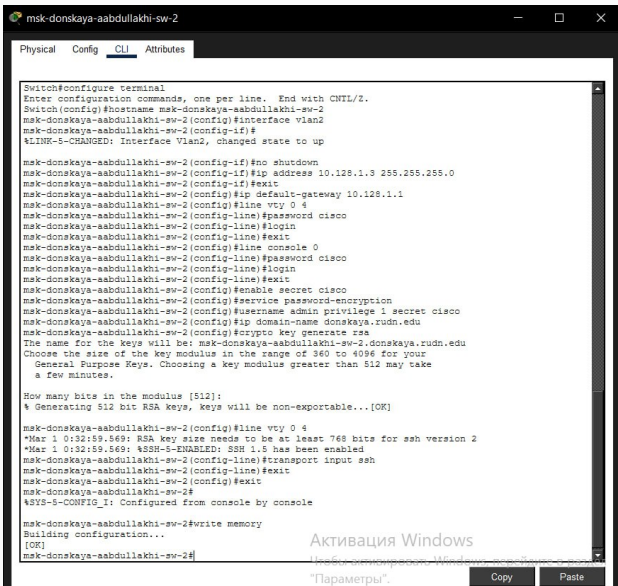
msk-donskaya-aabdullakhi-sw-1(config-if)#no shutdown
msk-donskaya-aabdullakhi-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-aabdullakhi-sw-1(config-if)#exit
msk-donskaya-aabdullakhi-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#exit
msk-donskaya-aabdullakhi-sw-1(config)#line console 0
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#exit
msk-donskaya-aabdullakhi-sw-1(config)#enable secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:22:43.551: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:22:43.555: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-1(config-line)#exit
msk-donskaya-aabdullakhi-sw-1(config)#exit
```

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Настройка msk-donskaya-aabdullakhi-sw-2



```
msk-donskaya-aabdullakhi-sw-2
Physical Config CLI Attributes

Switch#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Switch(config)#hostname msk-donskaya-aabdullakhi-sw-2
msk-donskaya-aabdullakhi-sw-2(config)#interface vlan2
msk-donskaya-aabdullakhi-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-aabdullakhi-sw-2(config-if)#no shutdown
msk-donskaya-aabdullakhi-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-aabdullakhi-sw-2(config-if)#exit
msk-donskaya-aabdullakhi-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-aabdullakhi-sw-2(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-2(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-2(config-line)#login
msk-donskaya-aabdullakhi-sw-2(config-line)#exit
msk-donskaya-aabdullakhi-sw-2(config)#line console 0
msk-donskaya-aabdullakhi-sw-2(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-2(config-line)#login
msk-donskaya-aabdullakhi-sw-2(config-line)#exit
msk-donskaya-aabdullakhi-sw-2(config)#enable secret cisco
msk-donskaya-aabdullakhi-sw-2(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-2(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-2.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

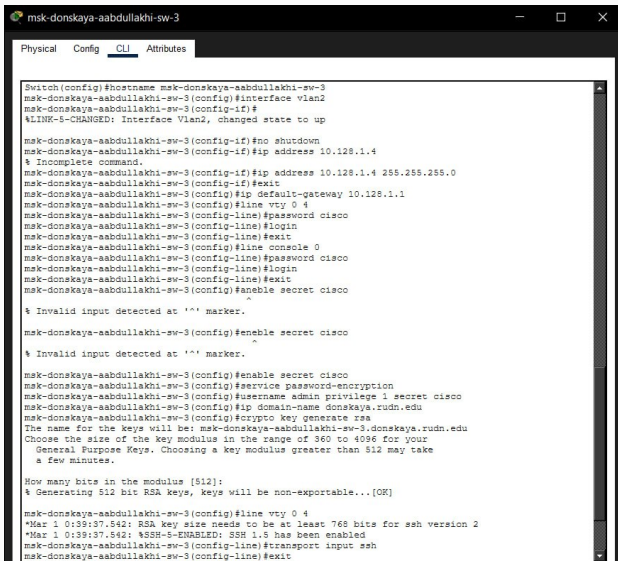
msk-donskaya-aabdullakhi-sw-2(config)#line vty 0 4
*Mar 1 0:32:59.569: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:32:59.569: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-2(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-2(config-line)#exit
msk-donskaya-aabdullakhi-sw-2(config)#exit
msk-donskaya-aabdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-aabdullakhi-sw-2#
```

Активация Windows
Используйте лицензионный ключ Windows или войдите в систему с помощью учетной записи Microsoft.
"Параметры".

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Настройка msk-donskaya-aabdullakhi-sw-3



```
Switch(config)#hostname msk-donskaya-aabdullakhi-sw-3
msk-donskaya-aabdullakhi-sw-3(config)#interface vlan2
msk-donskaya-aabdullakhi-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-aabdullakhi-sw-3(config-if)#no shutdown
msk-donskaya-aabdullakhi-sw-3(config-if)#ip address 10.128.1.4
% Incomplete command.
msk-donskaya-aabdullakhi-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-aabdullakhi-sw-3(config-if)#exit
msk-donskaya-aabdullakhi-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-aabdullakhi-sw-3(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-3(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-3(config-line)#login
msk-donskaya-aabdullakhi-sw-3(config-line)#exit
msk-donskaya-aabdullakhi-sw-3(config)#line console 0
msk-donskaya-aabdullakhi-sw-3(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-3(config-line)#login
msk-donskaya-aabdullakhi-sw-3(config-line)#exit
msk-donskaya-aabdullakhi-sw-3(config)#enable secret cisco
^
% Invalid input detected at '^' marker.

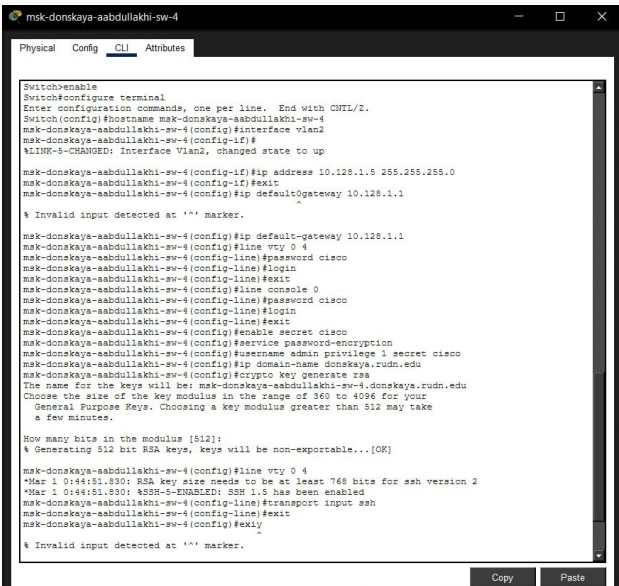
msk-donskaya-aabdullakhi-sw-3(config)#enable secret cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-3(config)#enable secret cisco
msk-donskaya-aabdullakhi-sw-3(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-3(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-3.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-sw-3(config)#line vty 0 4
*Mar 1 0:39:37.542: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:39:37.542: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-3(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-3(config-line)#exit
```

Настройка msk-donskaya-aabdullakhi-sw-4



```
msk-donskaya-aabdullakhi-sw-4
Physical Config CLI Attributes

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Switch(config)#hostname msk-donskaya-aabdullakhi-sw-4
msk-donskaya-aabdullakhi-sw-4(config)#interface vlan2
msk-donskaya-aabdullakhi-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

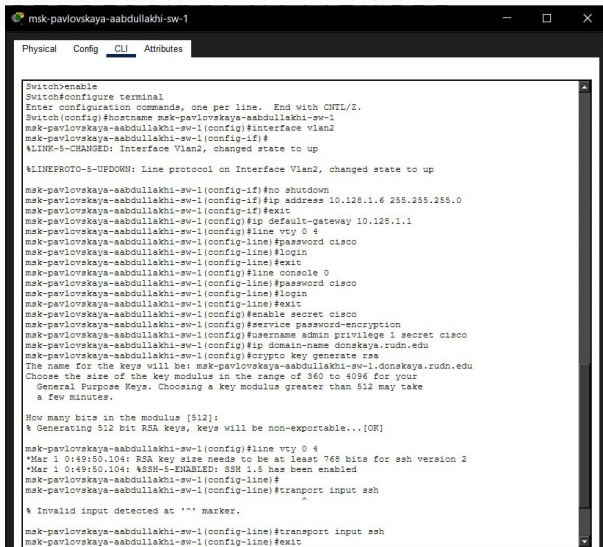
msk-donskaya-aabdullakhi-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-aabdullakhi-sw-4(config-if)#exit
msk-donskaya-aabdullakhi-sw-4(config)#ip default-gateway 10.128.1.1
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-aabdullakhi-sw-4(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-4(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-4(config-line)#login
msk-donskaya-aabdullakhi-sw-4(config-line)#exit
msk-donskaya-aabdullakhi-sw-4(config)#line console 0
msk-donskaya-aabdullakhi-sw-4(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-4(config-line)#login
msk-donskaya-aabdullakhi-sw-4(config-line)#exit
msk-donskaya-aabdullakhi-sw-4(config)#enable secret cisco
msk-donskaya-aabdullakhi-sw-4(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-4.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-sw-4(config)#line vty 0 4
*Mar 1 0:44:51.830: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:44:51.830: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-4(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-4(config-line)#exit
msk-donskaya-aabdullakhi-sw-4(config)#exit
^
% Invalid input detected at '^' marker.
```

Настройка msk-pavlovskaya-aabdullakhi-sw-1



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-pavlovskaya-aabdullakhi-sw-1
msk-pavlovskaya-aabdullakhi-sw-1(config)#interface vlan2
msk-pavlovskaya-aabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-pavlovskaya-aabdullakhi-sw-1(config-if)#no shutdown
msk-pavlovskaya-aabdullakhi-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-aabdullakhi-sw-1(config-if)#exit
msk-pavlovskaya-aabdullakhi-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-aabdullakhi-sw-1(config)#line vty 0 4
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#login
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-aabdullakhi-sw-1(config)#line console 0
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#login
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-aabdullakhi-sw-1(config)#enable secret cisco
msk-pavlovskaya-aabdullakhi-sw-1(config)#service password-encryption
msk-pavlovskaya-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-aabdullakhi-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-aabdullakhi-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:49:50.104: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:49:50.104: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#transport input ssh
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-aabdullakhi-sw-1(config-line)#transport input ssh
msk-pavlovskaya-aabdullakhi-sw-1(config-line)#exit
```

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Вывод

В ходе выполнения лабораторной работы: - смоделирована топология сети в соответствии с заданием; - выполнена начальная настройка пяти коммутаторов; - каждому устройству присвоен уникальный управляющий IP-адрес; - организована защита управляющего доступа с использованием локальных учётных записей и протокола SSH; - все изменения сохранены в энергонезависимую память.

Все коммутаторы готовы к эксплуатации и удалённому администрированию в пределах подсети **10.128.1.0/24**.